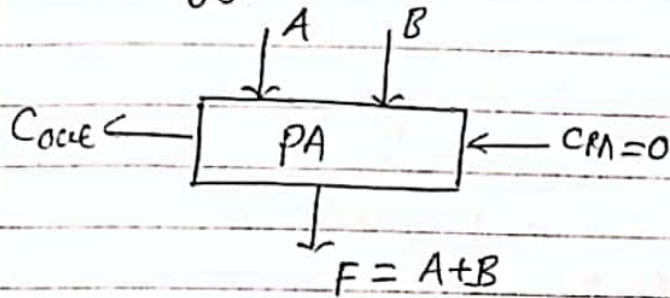
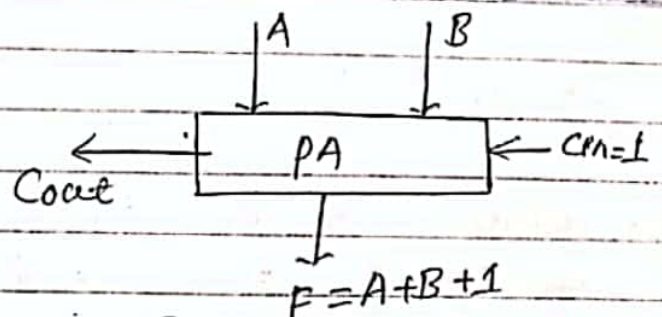


Design of Arithmetic Circuits:

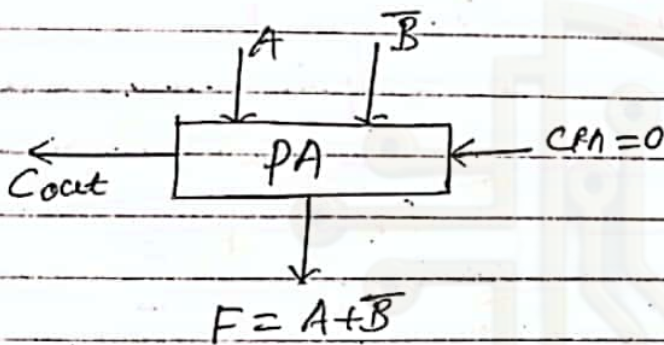
- The basic component of the arithmetic section of an ALU is a parallel adder. A parallel adder is constructed with a number of full adder connected in cascade.
- 5 different Arithmetic operations are:



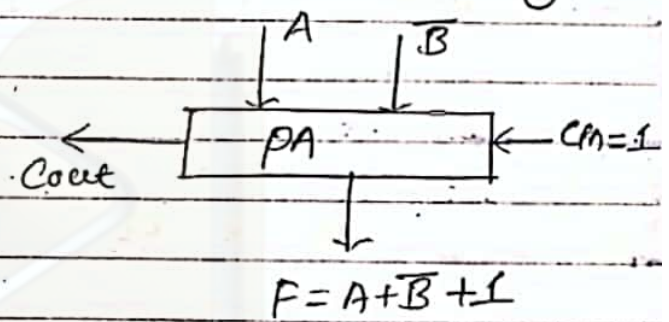
① Addition



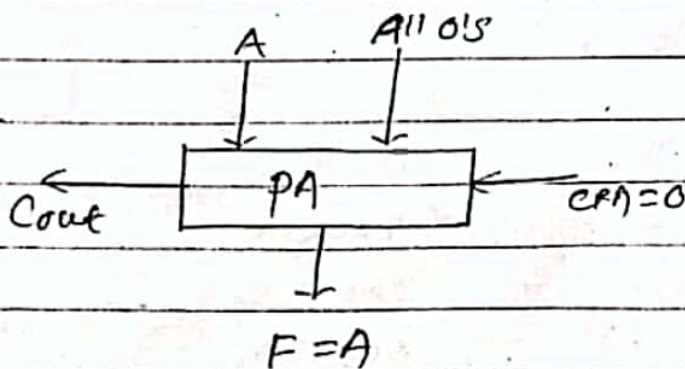
② Addition with carry.



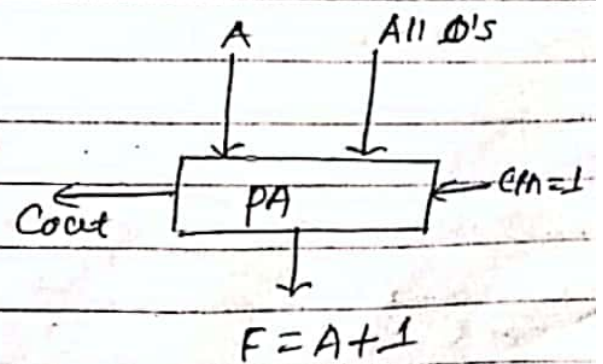
③ A plus 1's complement of B



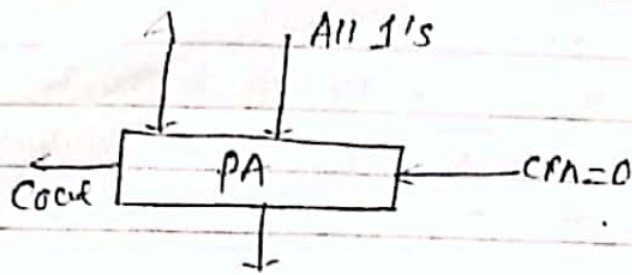
④ Subtraction



⑤ Transfer A

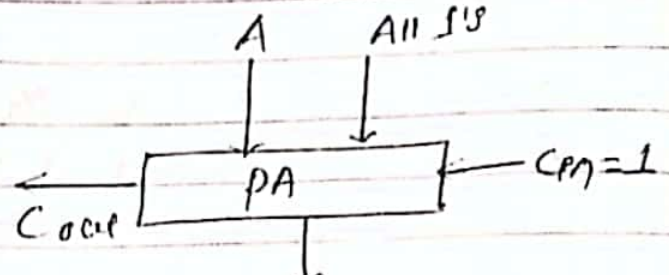


⑥ Increment A



$$F = A - 1$$

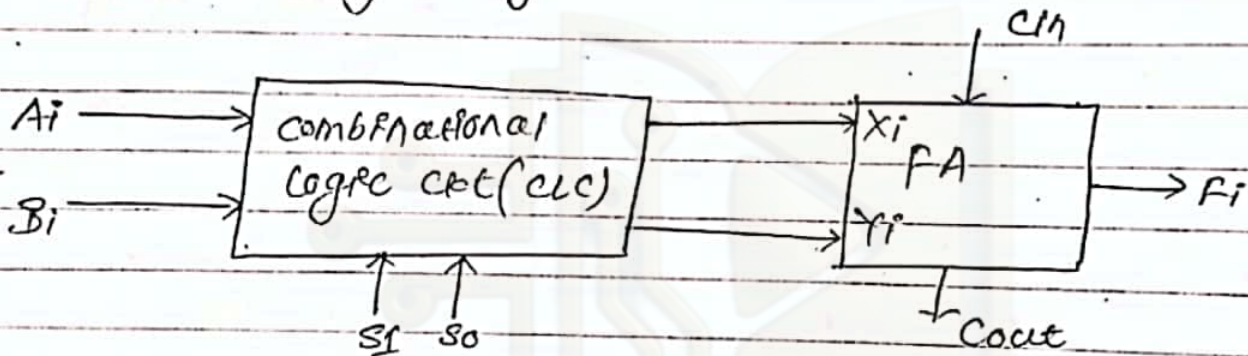
⑦ Decrement A



$$F = A$$

⑧ Transfer A

- Objective of arithmetic circuit design is to design a combinational circuit before the parallel adder, so that it can produce required Fp according to the select lines S_1 & S_0 .
- The block diagram of Arithmetic circuit is;



→ Function table:

Function select		Cin	F	Remarks
S_1	S_0			
→ 0	0	0	A	Transfer
0	0	1	$A + 1$	Increment
→ 0	1	0	$A + B$	Addition
0	1	1	$A + B + 1$	Addition with $Cin = 1$
→ 1	0	0	$A + \bar{B}$	$A + 1's \text{ comp. of } B$
1	0	1	$A + \bar{B} + 1$	$A - B$
→ 1	1	0	$A + \text{all } 1's$	$A - 1$
1	1	1	$A + \text{all } 1's + 1$	A

→ Truth table for CLC 18 08:

I/p's				O/p's	
S1	S0	Ai	Bi	X_i	Y_i
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	0	1	0
0	0	1	1	1	0
0	1	0	0	0	0
0	1	0	1	0	1
0	1	1	0	1	0
0	1	1	1	1	1
1	0	0	0	0	1
1	0	0	1	0	0
1	0	1	0	1	1
1	0	1	1	1	0
1	1	0	0	0	1
1	1	0	1	0	1
1	1	1	0	1	1
1	1	1	1	1	1

→ using K-Map for, X_i & Y_i

(i) for X_i

from above truth table, we can see that, ~~the value~~

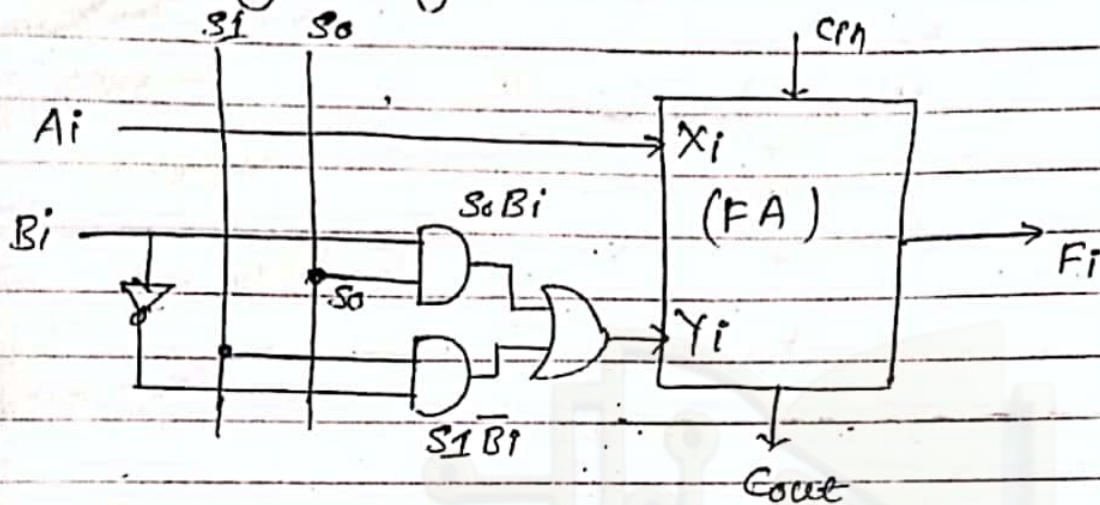
$$X_i = A_i$$

(ii) for Y_i

S1 \ S0	Ai Bi			
	00	01	11	10
00				
01		1	1	
11	1	1	1	1
10	1			1

$$\therefore Y_i = S_0 B_i + S_1 \bar{B}_i$$

→ The single stage arithmetic ckt is as;



→ Let the inputs A and B are both of 4-bit length, i.e.

$$A = A_4 A_3 A_2 A_1$$

$$B = B_4 B_3 B_2 B_1$$

then we draw the following logic diagram for ALU supporting various arithmetic operation on A & B.

→ logic diagram of

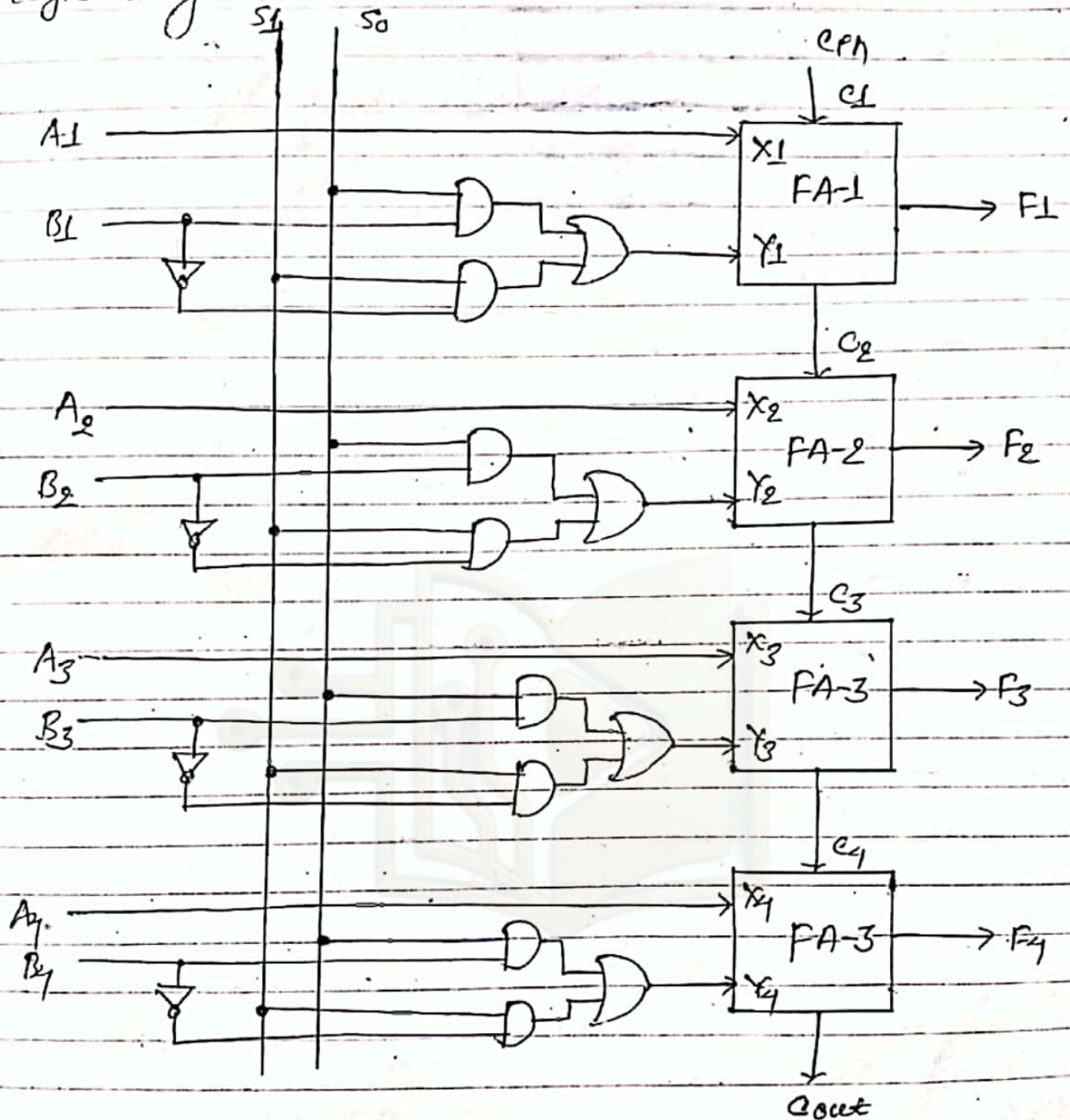


fig: 4-bit logic diagram of Arithmetic. ckt.

⑧ Design an arithmetic circuit with two selection variables s_1 and s_0 , that generate the following arithmetic operation. Draw the logic diagram of one typical stage.

s_1	s_0	$C_{in} = 0$	$C_{in} = 1$
0	0	$F = A + B$	$F = A + B + 1$
0	1	$F = A$	$F = A + 1$
1	0	$F = \overline{B}$	$F = \overline{B} + 1$
1	1	$F = A + \overline{B}$	$F = A + \overline{B} + 1$

A+B:

→ FUNCTION TABLE

S ₁	S ₀	C _{PA}	F
0	0	0	A+B
0	0	1	A+B+1
0	1	0	A
0	1	1	A+1
1	0	0	A B
1	0	1	B+1
1	1	0	A+B
1	1	1	A+B+1

→ Truth table

S ₁	S ₀	A _i	B _i	X _i	Y _i
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	1
0	1	0	0	0	0
0	1	0	1	0	0
0	1	1	0	1	0
0	1	1	1	1	0
1	0	0	0	0	1
1	0	0	1	0	0
1	0	1	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	0	1	0	0
1	1	1	0	1	1
1	1	1	1	1	0

→ K-Map

① for X_i

S ₁ S ₀ \ A _i B _i	00	01	11	10
00			1	1
01			1	1
11			1	1
10				

$$X_i = \bar{S}_1 A_i + \bar{S}_0 A_i$$

$$= A_i (\bar{S}_1 + \bar{S}_0)$$

② for Y_i

S ₁ S ₀ \ A _i B _i	00	01	11	10
00		1	1	
01				
11	1			1
10	1			1

$$Y_i = S_1 \bar{B}_i + \bar{S}_1 \bar{S}_0 B_i$$

→ One stage (single stage) Arithmetic logic diagram is,

$$X_i = A_i(\bar{S}_1 + \bar{S}_0)$$

$$Y_i = S_1 B_i + \bar{S}_1 \bar{S}_0 B_i$$

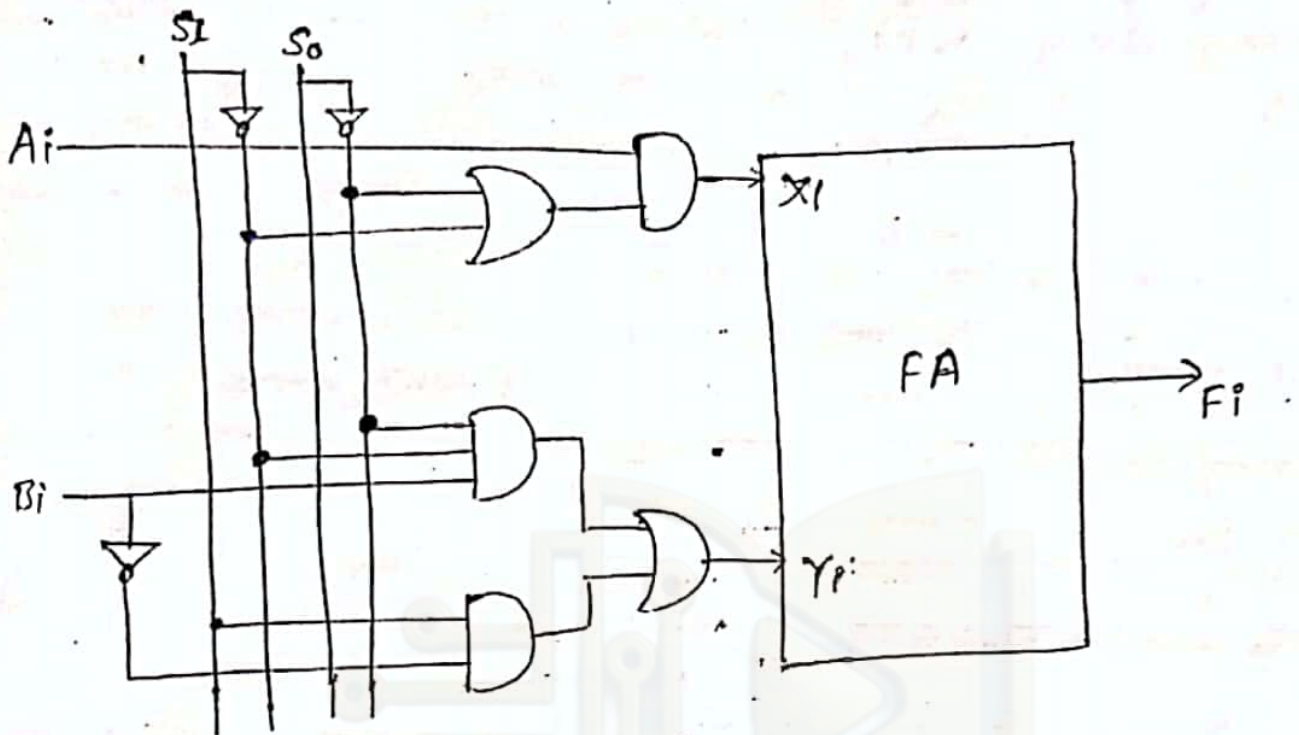


Fig: logic diagram of given arithmetic function table.

Q.5

Design an arithmetic. ckt with the two selection variables S_1 and S_0 that generates the following arithmetic operations. Draw the logic diagram of the typical stage.

S_1	S_0	$C_n = 0$	$C_n = 1$
0	0	$F = A$	$F = A + 1$
0	1	$F = A - B - 1$	$F = A - B$
1	0	$F = B - A - 1$	$F = B - A$
1	1	$F = A + B$	$F = A + B + 1$

Ans: $X_i = A_i(\bar{S}_1 + \bar{S}_0) + \bar{A}_i \bar{S}_1 \bar{S}_0$; $Y_i = B_i S_1 + \bar{B}_i \bar{S}_1 \bar{S}_0$

Status Register

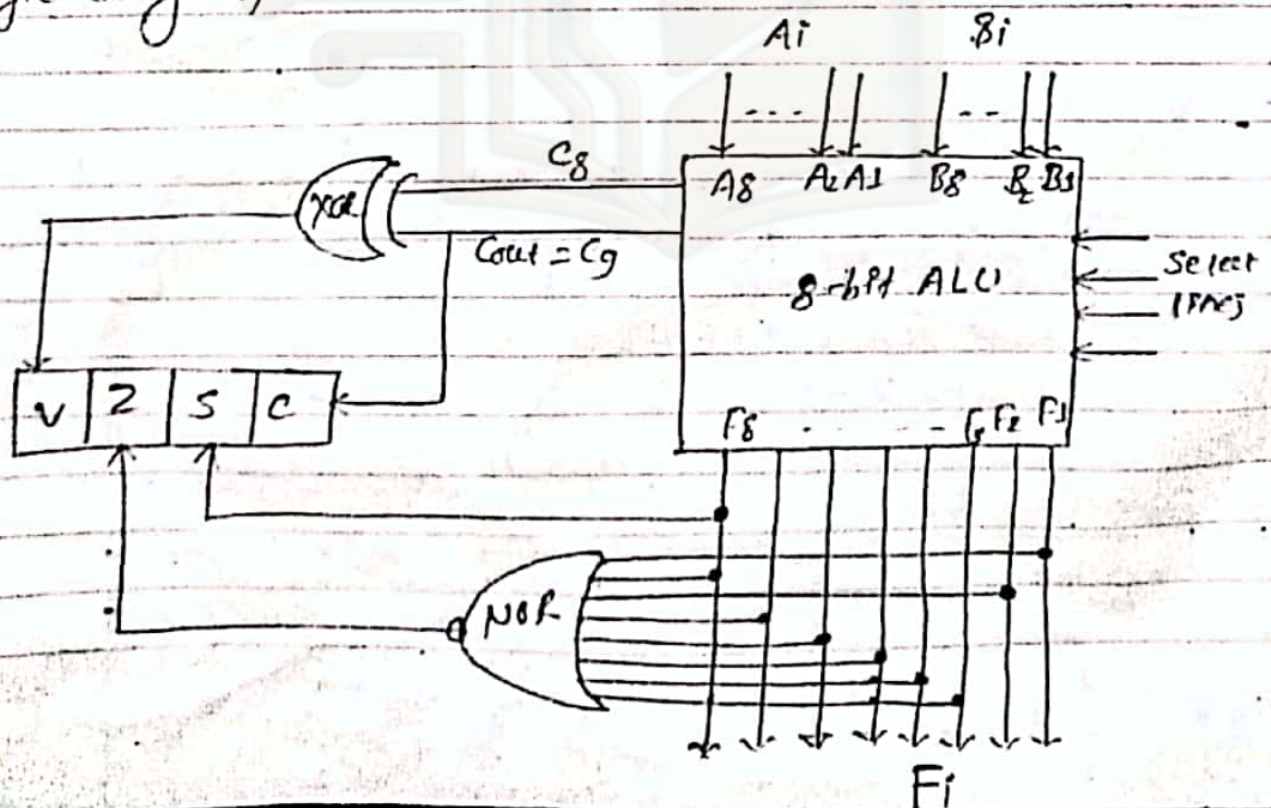
- Status Register (Flag Register) is a group of flip-flop used to give status of different operation result.
- It is connected to ALU.
- Once the operation is performed by ALU (Arithmetic Logic Unit), result is transferred on internal data bus and status of result will be stored in flip-flop.

→ 4 flag register are used

- ① Carry flag (CF) → CF is set if carry out of MSB.
- ② Sign flag (SF) → SF is set if result is negative.
- ③ Zero flag (ZF) → ZF is set if result is zero.
- ④ Overflow flag (OF) → OF is set if signed result is out of the range.

eg, for 8-bit ALU, OF (or V) is set if the result is greater than 127 and less than -128.

→ logic diagram



x/1-Flag

V $\rightarrow$ overflow

Z $\rightarrow$ zero

S $\rightarrow$ sign

C $\rightarrow$ carry

* working operation

① Bit C is set if the o/p carry of the ALU is 1. It is cleared if the o/p carry is 0.

② Bit S is set if the highest-order bit (MSB) of the result of the o/p of ALU is 1. It is cleared if the highest-order bit is 0.

③ Bit Z is set if the o/p of ALU contains all 0's, and cleared otherwise. i.e.

$Z = 1$ if the result is zero, and $Z = 0$ if the result is non-zero.

④ Bit V is set if the XOR of carries C_8 and C_9 is 1 otherwise V is cleared. This is the condition of overflow.

$\rightarrow$ states bits after the subtraction of unsigned numbers (A-B):

Relation	condition of status bit	Boolean function
$A > B$	$C = 1 \vee Z = 0$	CZ'
$A \geq B$	$C = 1$	C
$A < B$	$C = 0$	C'
$A \leq B$	$C = 0 \vee Z = 1$	$C' + Z$
$A = B$	$Z = 1$	Z
$A \neq B$	$Z = 0$	Z'

a) Design an arithmetic circuit with two selection variables S_1 and S_0 that generates the following arithmetic operations.

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S_1	S_0	$C_{in} = 0$	$C_{in} = 1$
0	0	$F = A$	$F = A + 1$
0	1	$F = A - B - 1$	$F = A - B$
1	0	$F = B - A - 1$	$F = B - A$
1	1	$F = A + B$	$F = A + B + 1$

b) Design a 3-bit synchronous gray code up counter using D flip-flop.

8

b) Design a 4-bit arithmetic circuits which performs eight different arithmetic operations.